

Review retrospective:
*the multivariate covariance theorem — a clean weak-track pass,
 a header that advertised the sharp result, and a docstring
 claiming sorries that weren't there*

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2026-06-05

Abstract

A soundness review of `Laplace/Multi/Covariance.lean` (3998 lines): the Stage-3 multivariate Laplace covariance theorem `lem:laplace_cov`, whose headline `gibbsCov_first_order_rate_weak` establishes $\text{Cov}_t[\varphi, \psi] = \frac{1}{t} \langle a, \Sigma b \rangle + o(1/t)$ (weak track, $\Sigma = H^{-1}$). Result: a clean fidelity pass at the statement level; the landables were two dead opens, three stale docstrings, and one dead private lemma. Two of the docstring fixes are the interesting part — a header that advertised the *sharp* rate this file does not prove, and a lemma docstring that claimed “internal sorries” in a proof that has none.

1 Setting

The second of the large `Multi/` analytic files, built on `RescaledIntegrals`, and the weak-track half of the multivariate covariance result (the sharp-track half is `gibbsCov_first_order_rate_sharp` in `CovarianceSharp.lean`). Roughly 2600 of its 3998 lines are private asymptotic-integral infrastructure between the public headlines.

2 What was reviewed

The five public headlines — the partition asymptotic `rescaledPartition_eq_gaussianZ_add_0_inv_sqrt` and its lower bound, the single-observable bound, the pair expectation `rescaledExpectation_pair_eq_main_add_0_inv_sqrt`, and `gibbsCov_first_order_rate_weak` — checked for the $1/t$ rate, the $\langle a, H^{-1}b \rangle$ main contraction, the Gaussian sign, and the error orders.

3 Fidelity / soundness findings

Sound at the statement level. The reviewer confirmed every headline: the main term contracts through H^{-1} (not H), the Gaussian pair term has the right sign, and the scaled errors are $O(t^{-1/2})$ (so $\text{Cov}_t = \text{main}/t + O(t^{-3/2}) = o(1/t)$). “No headline-level H vs H^{-1} mistake, no sign error.”

Three stale/over-claiming docstrings, two of them notable.

- *The header advertised the sharp result.* The module header stated $\text{Cov}_t = \frac{1}{t} \langle a, \Sigma b \rangle + O(t^{-2})$ and said the file “(eventually) proves” the explicit $\leq K/t$ bound — but the public theorem is

named `_weak` and proves $\leq K/\sqrt{t}$. The sharp $O(t^{-2})$ result is a *different, deliberately separate* theorem (`gibbsCov_first_order_rate_sharp` in `CovarianceSharp`, whose own header cross-references back here). So this is not an under-proven theorem to escalate — it is a header describing the wrong file’s result. Fixed to state the weak form and cross-reference the sharp track. (The first attempt fixed the body paragraph but left the opening line still reading $O(t^{-2})$; the affirmation pass caught that, and a second pass corrected the opening line to $o(1/t)$.)

- A docstring claimed *sorries that do not exist*. The private `abs_integral_dot_mul_rescaled_weight_correction.le` was titled “(in-progress, contains internal sorries)” and said two integrability facts “remain as internal sorries” — but a `grep`, a green build, and `scripts/sorries` all confirm the file has *zero* sorries. The proof was completed and the docstring never updated; a reader would wrongly think the result unsound. Corrected to state the (completed) derivation.
- The headline’s own docstring still said “statement only / proof in progress” (it is proved) and pointed at a non-existent filename `Covariance.Sharp.lean`; both corrected.

4 Simplifications applied

Two dead opens and a dead private lemma. `open Module` was fully dead (no `Module.*`/`finrank` anywhere); the section-local `open MeasureTheory` was redundant under the namespace-level `open` (the same pattern as `RescaledIntegrals`). And the private `integrable_dot_dot_mul_rescaled_weight_correction` had no call sites — the bilinear correction re-proves its integrability inline — so it was removed (private, so the public surface is untouched). Build green, public signatures byte-identical.

5 Disagreements and open questions

None left open. The affirmation pass rejected the first header fix as incomplete (the opening line still advertised $O(t^{-2})$); the second pass resolved it. Several further cleanups are report-only for a hygiene tide: the `*_def` dead equation-names and dead-local constant blocks (on this `positivity/field_simp`-heavy file they carry the implicit-consumption risk, so they want individual build-arbitration), and a few unused `hGauss` parameters on private lemmas (a signature change of private decls, multi-site, low value).

6 Lessons

- **A header can advertise a sibling file’s result.** When a result is split into a weak-track and a sharp-track file, the weak file’s header is prone to stating the *sharp* theorem as its own headline. The tell: the public theorem is named `_weak` but the prose claims the sharp rate. The fix is to scope the header to what the file proves and cross-reference the other track — not to escalate (the theorem is complete, just mislabelled).
- **“Contains internal sorries” is a verifiable claim — verify it.** A docstring advertising sorries in a file that has none is worse than a mere staleness nit: it impugns a sound result. `grep` + green build + `scripts/sorries` settle it in seconds; here all three agreed the lemma is complete.

- **The affirmation pass catches half-done doc fixes.** My header fix corrected the body but left the opening line over-claiming; sending the diff back caught the residue. A docstring fix is not done until *every* line of the docstring matches the theorem.

7 Follow-ups

- Hygiene tide: build-arbitrate the `*_def` / dead-local-block candidates and the unused private `hGauss` parameters.
- The last never-reviewed laplace files are `CovarianceSharp` (4179L) and `CovarianceExplicit` (19611L) — the latter genuinely sectioned-review territory.